



Tachiwin

PaddleOCR Global Derivative Model Challenge — Hackathon 10th

Training Data Construction Report

Tachiwin-OCR-1.5: Indigenous Languages of Mexico

This report documents the training dataset construction methodology, fine-tuning procedure, evaluation dataset design, and benchmark results for Tachiwin-OCR-1.5 — a fine-tuned derivative of PaddleOCR-VL-1.5 specialized in the 68 officially recognized indigenous languages of Mexico (INALI).

68	+33	14.5 GB	49K	33K
New language groups covered	Specialized characters	Training dataset	Training samples	Evaluation dataset

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Project Overview

Tachiwin-OCR-1.5 is a fine-tuned derivative of PaddleOCR-VL-1.5 specialized in the 68 officially recognized indigenous languages of Mexico as defined by the Instituto Nacional de Lenguas Indígenas (INALI). These languages and their hundreds of orthographic variants are spoken by over 7 million people, yet have no dedicated OCR solution in either academia or industry.

The project addresses a specific and well-defined failure mode: standard OCR models have never encountered the orthographic characters used systematically across this language family. Specialized latin script characters and combining tone marks, appear constantly in indigenous language text but are out-of-distribution for any general-purpose OCR model trained primarily on Spanish, English, and other latin-script high-resource languages.

The project covers the full pipeline: construction of a large synthetic training dataset derived from real indigenous language corpora, supervised fine-tuning of PaddleOCR-VL-1.5, construction of a real-PDF evaluation dataset with no synthetic data, and rigorous benchmarking with statistical validation.

Model: tachiwin/Tachiwin-OCR-1.5 | Base: PaddlePaddle/PaddleOCR-VL-1.5 | License: Apache 2.0

1. Training Dataset Construction

The training dataset (tachiwin/multilingual_ocr_llm_2, ~14.5 GB, 49,689 samples) was produced through a three-stage pipeline: language index assembly, corpus collection and cleaning, and synthetic image rendering.

1.1 Language index

A structured language index was assembled covering all 68 INALI-recognized indigenous language groups and their documented variants, each tagged with its ISO 639-3 analogue code. This index served as the authoritative reference for both corpus collection and character set derivation, ensuring systematic coverage across the full language family rather than ad-hoc selection.

1.2 Corpus collection and cleaning

Using the language index, a broad web scraping pipeline collected a large corpus of indigenous language text from primarily governmental and institutional websites (including INALI, SEP, and regional indigenous affairs offices). Text was categorized per language using the ISO index.

The collected text was then cleaned and prepared for rendering:

- Malformed UTF-8 entities removed and all text NFC-normalized
- Text segmented into paragraphs with normally distributed paragraph length — ensuring the model sees a balanced range of short and long text blocks

1.3 Synthetic image rendering

Each paragraph was rendered as a training image using pseudorandom parameter sampling with normally distributed values around realistic medians. 31 curated font families (which fully covered

our specialized character set) were used across the full dataset. The following parameter space was sampled independently per image:

Group	Parameters	Distribution
Text appearance	Point size, char spacing, line spacing, uppercase	ptsize median 21 $\pm$ 11 pt; char_spacing $\pm$ 0.3; line_spacing median 0 $\pm$ 3
Color	Background gray, text gray, color mode, RGB jitter	Background mostly white (median 255 $\pm$ 80); text mostly dark (median 0 $\pm$ 80); ~25% chance of full RGB color with aggressive channel variance
Geometry	Short side, margin factor, aspect ratio	Short side median 500 $\pm$ 150 px; 60% horizontal (ratio 1.2–2.5), 20% vertical (0.7–0.8), 20% square (0.8–1.2)
Geometric distortions	Rotation, skew X/Y, perspective, wave	Rotation $\pm$ 0.5°; skew X/Y $\pm$ 0.5; perspective $\pm$ 0.1; wave ~30% chance (amplitude $\pm$ 0.5, freq $\pm$ 50)
Blur	Gaussian blur; motion blur (radius, sigma, angle)	Gaussian median 0.2 $\pm$ 1.0 σ ; motion blur ~30% chance, radius 4 $\pm$ 2, any angle 0–360°
Noise	Type: Gaussian / Impulse / Laplacian / Poisson / Uniform	~30% chance; Gaussian 30%, Impulse 30%, Laplacian 10%, Poisson 10%, Uniform 20%
Photometric	Brightness, contrast, dirty/stain overlay	Brightness $\pm$ 10, contrast $\pm$ 10; dirty overlay ~30% chance
Morphology	Erode/Dilate — kernel: Diamond, Disk, or Square	Simulates ink spread or character erosion
Text alignment	Left, center, right	60% left, 30% center, 10% right

With 31 font families combined with independently sampled continuous distortion parameters, the practical number of visually distinct configurations is in the tens of millions — producing an unbiased dataset that generalizes across the full spectrum of real-world text capture conditions: clean prints, scans, photographs, photocopies, and degraded or aged documents.

1.4 Character set derivation

The 33-character UNCOMMON_CHARS set — used both in training quality filtering and evaluation difficulty scoring — was derived through a corpus-driven lexicostatistical pipeline, not selected manually:

- Character frequency distributions were computed from corpora for each of the 68 language groups and their variants
- Characters already present in Spanish (es-MX) were excluded — Spanish serves as the canonical baseline to avoid disrupting existing OCR performance on the dominant language
- The differential character sets from all 68 languages were unioned into a single comprehensive set

The resulting 33 markers are characters that are linguistically significant across the indigenous language family but systematically absent from standard OCR training data. This character set was originally developed for a keyboard layout optimization algorithm (separately released

open-source) and repurposed here as a principled linguistic coverage and difficulty metric.

Character	Unicode	Name	Linguistic function
ɬ	U+019A	Latin Small Letter L with Bar	Lateral consonant variant
ʔ	U+A78C	Latin Small Letter Saltillo	Glottal stop / saltillo marker
ʌ	U+028C	Latin Small Letter Turned V	Open-mid back unrounded vowel
ʊ̯	U+0289	Latin Small Letter U Bar	Close central rounded vowel
ɛ	U+025B	Latin Small Letter Open E	Open-mid front vowel
ɪ̯	U+0268	Latin Small Letter I with Stroke	Close central vowel
ⁿ	U+207F	Superscript Latin Small Letter N	Prenasalization marker
-	U+2011	Non-Breaking Hyphen	Morpheme boundary marker
ˊ	U+02CA	Modifier Letter Acute Accent	Standalone rising tone diacritic
ˋ	U+02CB	Modifier Letter Grave Accent	Standalone falling tone diacritic
ɭ	U+1E3B	Latin Small Letter L with Line Below	Retroflex lateral
ɳ	U+1E49	Latin Small Letter N with Line Below	Retroflex nasal
ǔ	U+01D4	Latin Small Letter U with Caron	Tone marker (rising)
ǎ	U+01CE	Latin Small Letter A with Caron	Tone marker (rising)
ə	U+0259	Latin Small Letter Schwa	Mid-central vowel
ʼ	U+02BC	Modifier Letter Apostrophe	Saltillo / glottal stop
ˉ	U+02C9	Modifier Letter Macron	Level tone diacritic
ó	U+01FF	Latin Small Letter O with Stroke and Acute	Complex diacritic combination
ŋ	U+014B	Latin Small Letter Eng	Velar nasal
ɨ̯	U+012F	Latin Small Letter I with Ogonek	Nasalized vowel
ō	U+014D	Latin Small Letter O with Macron	Long/level-tone vowel
ā	U+0101	Latin Small Letter A with Macron	Long/level-tone vowel
ī	U+012B	Latin Small Letter I with Macron	Long/level-tone vowel
ē	U+0113	Latin Small Letter E with Macron	Long/level-tone vowel
ū	U+016B	Latin Small Letter U with Macron	Long/level-tone vowel
ž	U+017E	Latin Small Letter Z with Caron	Postalveolar fricative
š	U+0161	Latin Small Letter S with Caron	Postalveolar fricative

◌̣	U+0331	Combining Macron Below	Tone / length mark below
◌̣̣	U+0328	Combining Ogonek	Nasalization mark
◌̣̣̣	U+0304	Combining Macron	Length / level tone
◌̣̣̣̣	U+0308	Combining Diaeresis	Vowel quality mark
◌̣̣̣̣̣	U+0303	Combining Tilde	Nasalization mark
◌̣̣̣̣̣̣	U+0301	Combining Acute Accent	Rising tone / stress mark

Table: Complete UNCOMMON_CHARS set — 33 orthographic markers covering the indigenous language family of Mexico.

Training vs evaluation data: The training dataset uses synthetic image rendering to maximize volume. The evaluation dataset (tachiwin/ocr-test-challenging-3) is built exclusively from real PDF pages — no synthetic data. This separation ensures the benchmark reflects real-world performance, not training distribution memorization.

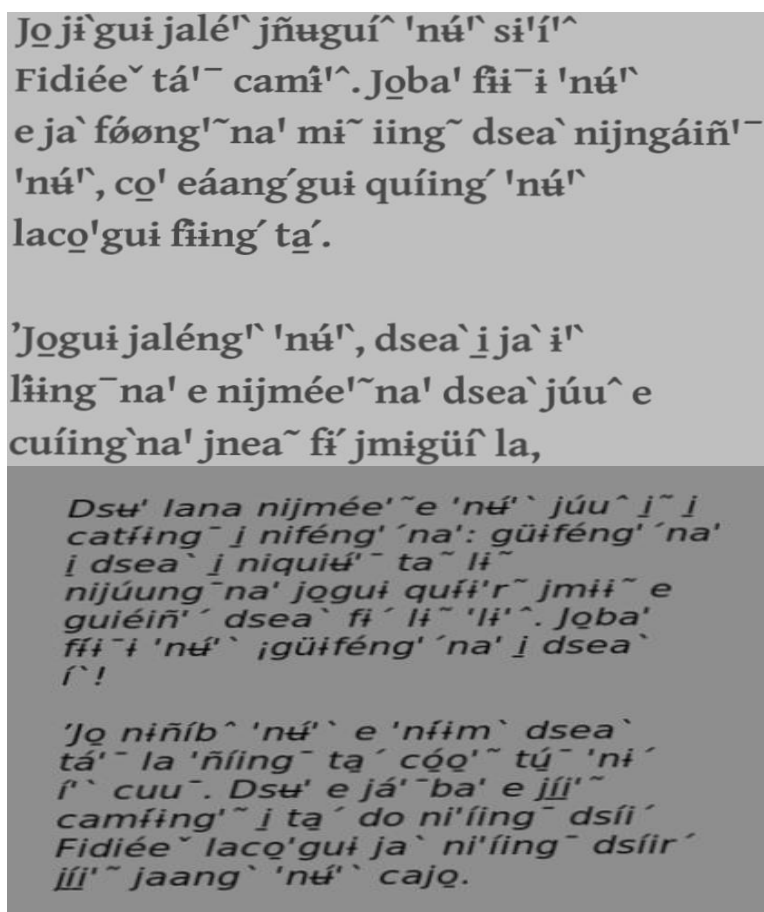


Figure 1. Sample training images with complex diacritics and specialized characters

2. Fine-Tuning

Fine-tuning was performed using Unsloth with the TRL SFTTrainer. As per standard PaddleOCR-VL fine-tuning practice, training targets the OCR model weights; the layout detection stage is language-agnostic and was left unchanged.

2.1 Training configuration

Parameter	Value	Notes
Base model	PaddlePaddle/PaddleOCR-VL-1.5	
Framework	Unsloth + TRL SFTTrainer	~2× faster vs standard HF training
Fine-tuning type	Full parameter SFT	full_finetuning=True
Training samples	49,689	tachiwin/multilingual_ocr_llm_2
Epochs	2	
Batch size	4	Per device
Gradient accumulation	8 steps	Effective batch size = 32
Learning rate	5e-5	
LR scheduler	Cosine	
Warmup ratio	0.05	155 steps
Optimizer	AdamW 8-bit	Memory-efficient
Weight decay	0.001	
Max sequence length	2,048 tokens	Padded to multiples of 8
Precision	bf16 / fp16	Auto-detected per GPU
Seed	3407	Reproducibility
Checkpointing	Every 500 steps HF Hub	save_total_limit=1; resumable
Experiment tracking	Weights & Biases	Run: tachiwin_ocr_1_5_a

2.2 Total training steps

```
total_steps = ceil(49689 / (4 × 8)) × 2 epochs = ceil(49689 / 32) × 2 = 1553 × 2 = 3,106 steps
warmup_steps = round(0.05 × 3106) = 155 steps
```

2.3 Training strategy notes

- Full fine-tuning over LoRA: All model parameters are updated. For a task requiring the model to learn an entirely new character vocabulary across 68 language groups, full fine-tuning provides greater capacity to adapt embedding and attention layers that LoRA would leave frozen.

- Response-only training: Only the assistant response tokens contribute to the loss. Prompt tokens (image + 'OCR:' instruction) are masked with `-100` using `UnslothVisionDataCollator` with `train_on_responses_only=True`, preventing the model from wasting capacity reproducing the fixed prompt.
- Resumable across sessions: A custom `ColabTimeoutCallback` detects approaching session timeouts, saves a checkpoint to Hugging Face Hub, and resumes exactly from that checkpoint in the next session — making multi-session training practical.

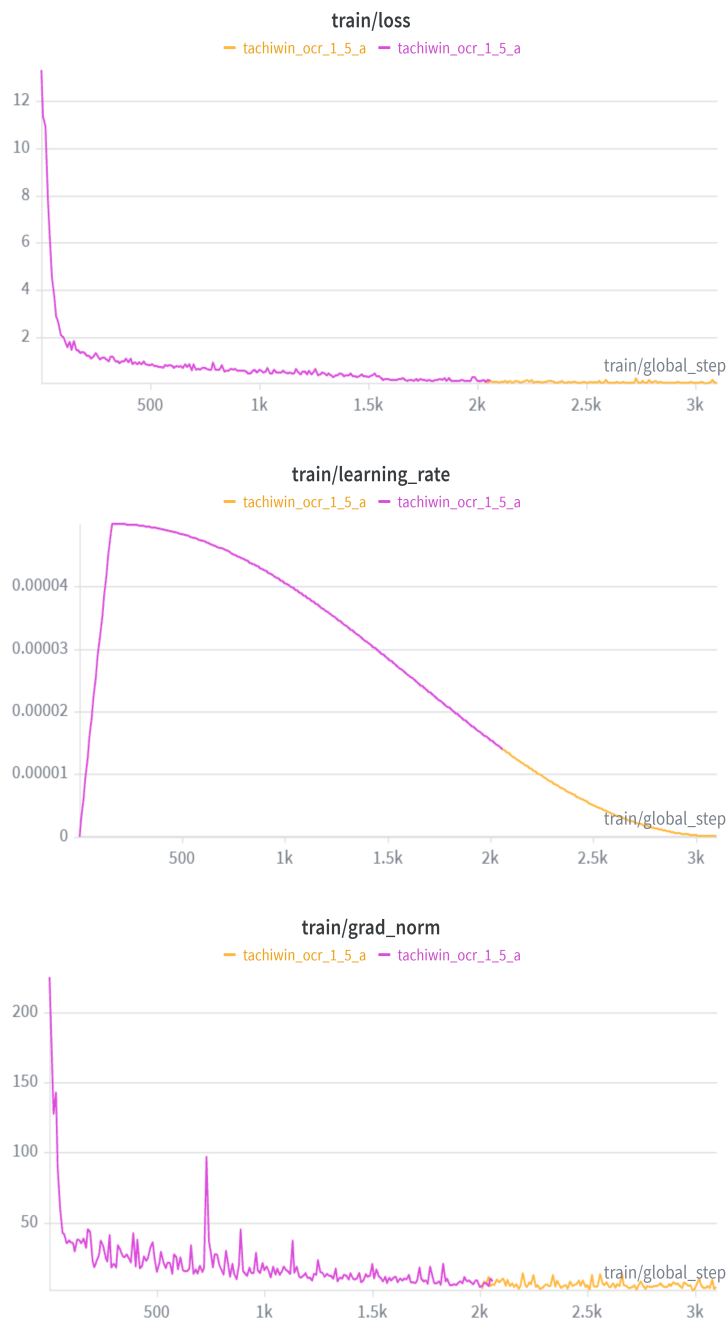


Figure 2. Finetune charts for train loss, learning rate and gradient norm.

3. Evaluation Dataset Construction

The evaluation dataset (tachiwin/ocr-test-challenging-3, ~10.5 GB, 33,000 rows) is built exclusively from real PDF documents, with no synthetic data. The construction pipeline is fundamentally different from the training pipeline.

3.1 Page selection criteria

Pages were drawn from a corpus of indigenous language PDF documents. A page was selected for the evaluation set only when it passed all of the following criteria simultaneously:

- Clean native text extraction: `pymupdf4llm` must extract text from the page with less than 2% replacement/box characters (U+FFFD, U+25A1), with a minimum character count of 100. This threshold ensures the extracted text is a reliable ground truth — not a partial or garbled extraction.
- Indigenous character density: The page must have an `uncommon_char_score` ≥ 0.3 (see below; configurable via `UNCOMMON_CHAR_SCORE_MIN`). This ensures every evaluation page contains a meaningful presence of indigenous-specific characters — the exact characters the model is designed to handle.
- Valid image render: The page must render successfully as a high-resolution PNG from the same PDF, with no blank or corrupt output.

3.2 Ground truth and test input

For each selected page, two outputs are generated from the same PDF page:

- Reference (ground truth): Text extracted by `pymupdf4llm` in markdown format, with HTML/image entities stripped.
- Test input: The page rendered as a PNG image, fed through the full PaddleOCR-VL pipeline (layout detection OCR markdown). HTML/image entities are also stripped from the OCR output before comparison.

Both the base model and the fine-tuned model run the identical PaddleOCR-VL pipeline — same layout detector, same markdown renderer. The only variable is the OCR model weights. Any performance difference is therefore attributable exclusively to OCR recognition quality.

3.3 uncommon_char_score

Each row in the dataset has a computed `uncommon_char_score` — a continuous value between 0.0 and 1.0 based on the log-scaled density and diversity of `UNCOMMON_CHARS` characters in the reference text:

```
score = log10(uncommon_count + 1) / log10(total_chars + 1) + min(0.3,
unique_uncommon_chars * 0.05) # Capped at 1.0. Higher = denser indigenous character
content.
```

The full 33,000-row dataset contains rows across the entire score range 0.0–1.0, reflecting the natural variation in indigenous character density across the document corpus. When running the evaluation script, an arbitrary minimum score threshold can be set via `UNCOMMON_CHAR_SCORE_MIN` to select any desired subset of the data. Setting it to 0.0 evaluates the entire unfiltered dataset; raising it restricts evaluation to pages with progressively higher indigenous character density. This makes it possible to run the evaluation at any desired level of character-complexity, as

required by the competition's variable-difficulty criteria.

For the benchmark reported in this submission, `UNCOMMON_CHAR_SCORE_MIN` = 0.3 was used, with a 2,000-item subset providing broad coverage across the score range. Results broken down by score range are reported in Section 4.

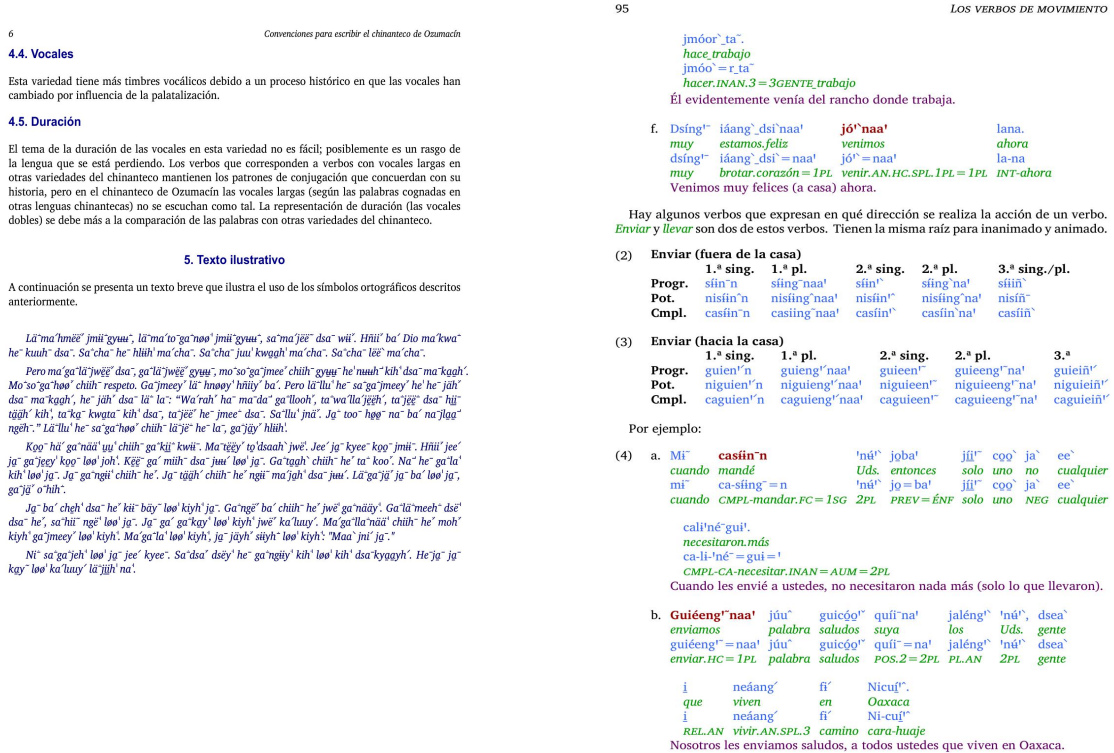


Figure 3. Samples from the evaluation dataset of two PDF pages containing `uncommon_char_score` 1.0 (the highest) containing a large amount of specialized characters which also have perfect text transcription

4. Evaluation Methodology and Results

4.1 Evaluation design

The evaluation script (`tachiwin_ocr_comparison_eval.py`, attached to this submission) performs a full paired comparison between the base model and Tachiwin-OCR-1.5 on the challenging evaluation dataset. It is fully self-contained:

- Downloads both models from Hugging Face Hub
- Launches a vLLM server as a subprocess
- Runs the evaluation loop with timeout and retry handling
- Computes CER, WER, char accuracy, and word accuracy per item
- Performs Wilcoxon signed-rank test and paired t-test for statistical validation
- Outputs per-item JSON and a summary report

Full-pipeline evaluation: Both models run the complete PaddleOCR-VL pipeline (layout detection OCR markdown reconstruction) on each page image. The pipeline constants — layout detector, reading order algorithm, markdown renderer — are shared and identical between both runs. Differences in CER/WER are therefore entirely attributable to the OCR model weights. Evaluating at the full-page level (rather than pre-segmented crops) reflects real-world deployment conditions and is a more demanding, realistic benchmark.

Paired design: Both models are evaluated on exactly the same pages, enabling paired statistical tests. HTML and image entities are stripped from both reference and OCR output before metric computation, so evaluation measures text recognition quality only.

```
# Key parameters in tachiwin_ocr_comparison_eval.py MAX_EVAL_ITEMS = 2000 # set to
None to run the full 33k dataset UNCOMMON_CHAR_SCORE_MIN = 0.3 # set to 0.0 for
unfiltered; raise for harder subsets # Dependencies vllm==0.11.1
paddlepaddle-gpu==3.4.1 paddleocr[doc-parser]==3.4.1 openai jiwer datasets scipy
huggingface_hub Pillow tqdm
```

The full 33,000-row dataset can be evaluated by setting `MAX_EVAL_ITEMS = None`. `UNCOMMON_CHAR_SCORE_MIN` can be set to any value between 0.0 and 1.0 to select subsets by indigenous character density — from the full unfiltered corpus (0.0) to pages with maximum character complexity (approaching 1.0). The 2,000-item benchmark subset runs in approximately 4–6 hours on a single A100.

4.2 Metrics

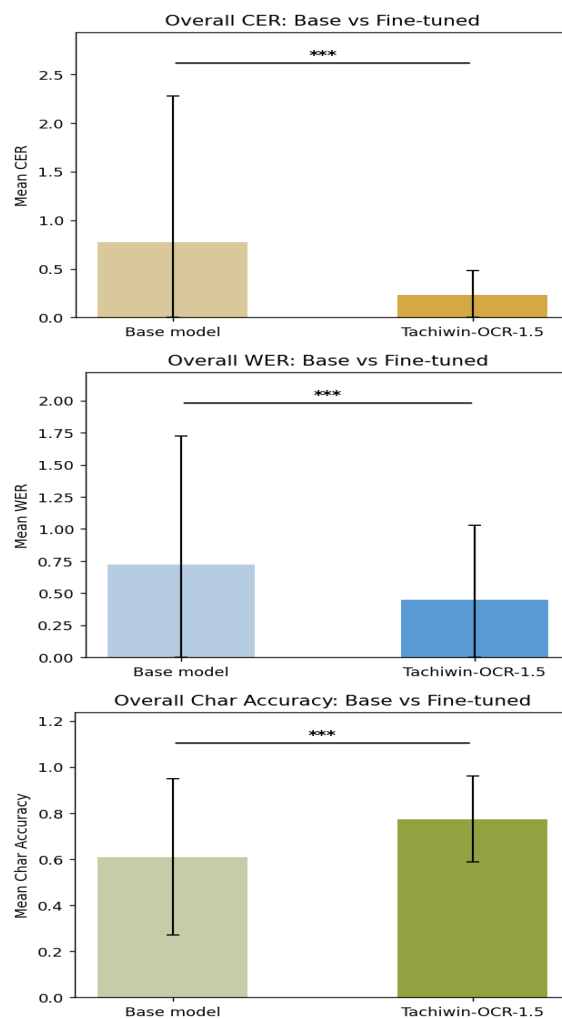
CER (Character Error Rate) is the primary metric. It operates at the character level and is less sensitive to the agglutinative morphology common in indigenous languages, where a single character error at a morpheme boundary can inflate word-level error counts disproportionately. WER, char accuracy, and word accuracy are reported as secondary metrics.

Statistical significance is assessed using the Wilcoxon signed-rank test (non-parametric, appropriate for paired samples with non-normal distributions). A paired t-test is computed as a secondary check.

4.3 Results

Results on the 2,000-item benchmark subset (uncommon_char_score 0.3, n=2,000 paired samples):

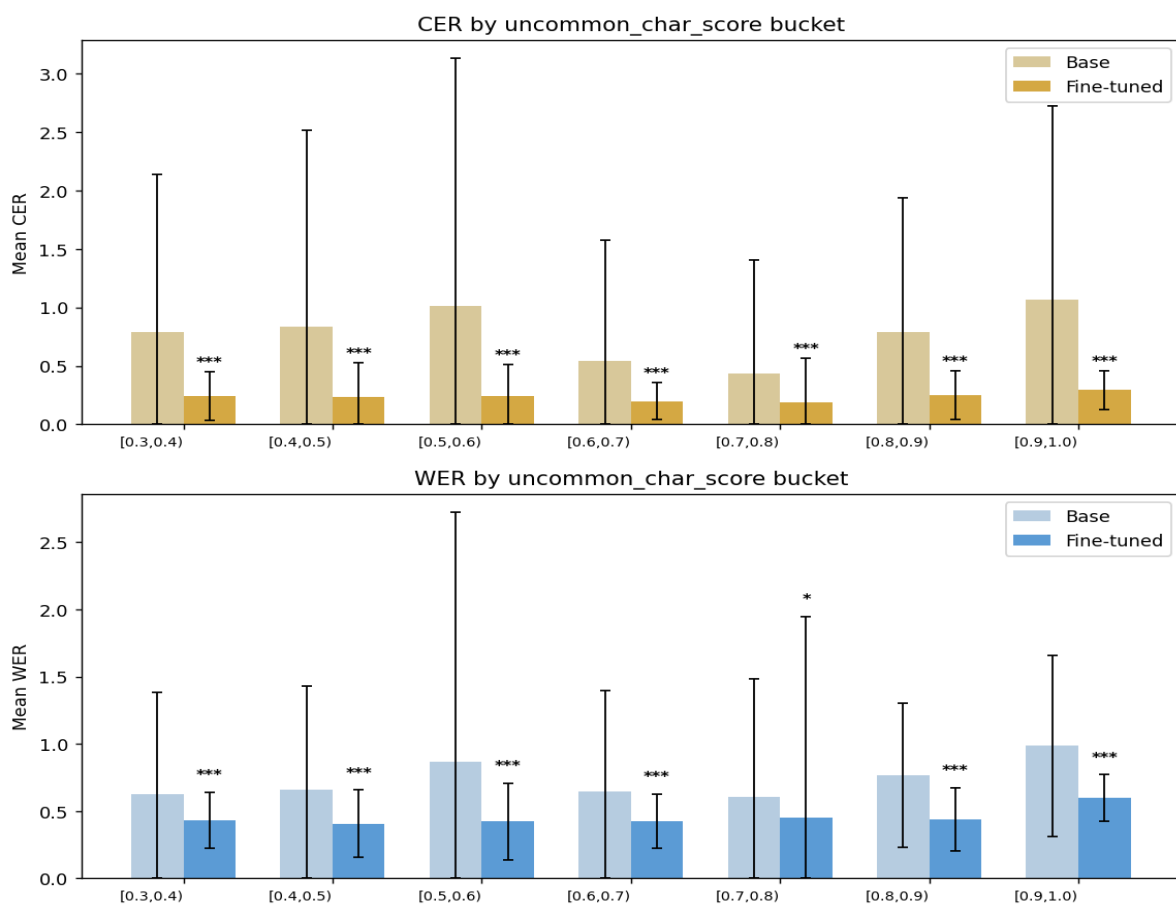
Metric	Base model	Tachiwin-OCR-1.5	Absolute Δ	Relative Δ	Wilcoxon p
CER	0.773	0.232	-0.541	-70%	< 0.0001 ***
WER	0.725	0.449	-0.276	-38%	< 0.0001 ***
Char accuracy	61.0%	77.4%	+16.4pp	+27%	< 0.0001 ***
Word accuracy	44.6%	56.6%	+12.0pp	+27%	< 0.0001 ***

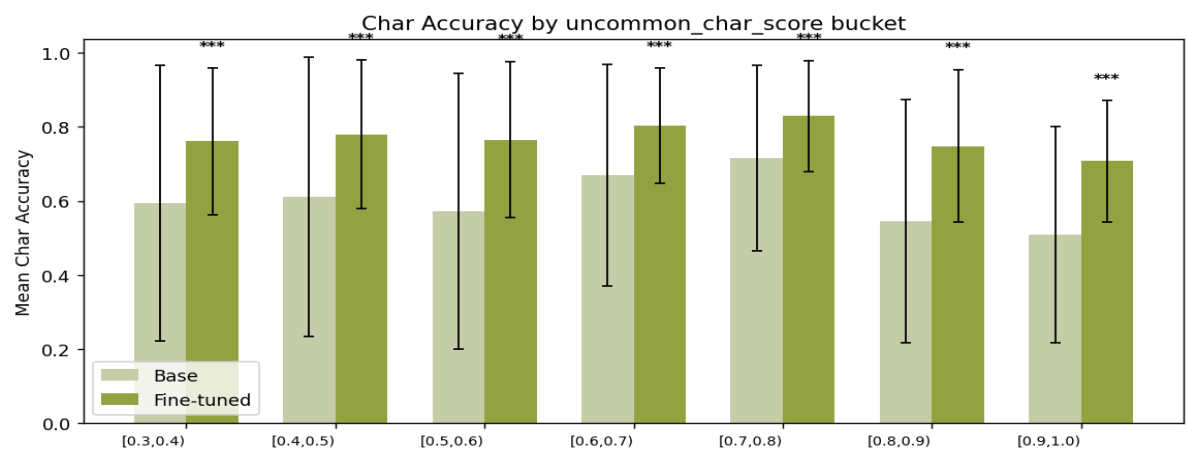


4.4 CER by score range

For analysis, the continuous `uncommon_char_score` of each evaluated page is discretized into ranges to show how performance varies with indigenous character density. These are not separate evaluation runs or pre-defined strata — they are a tabular representation of the distribution of results across the continuous score axis:

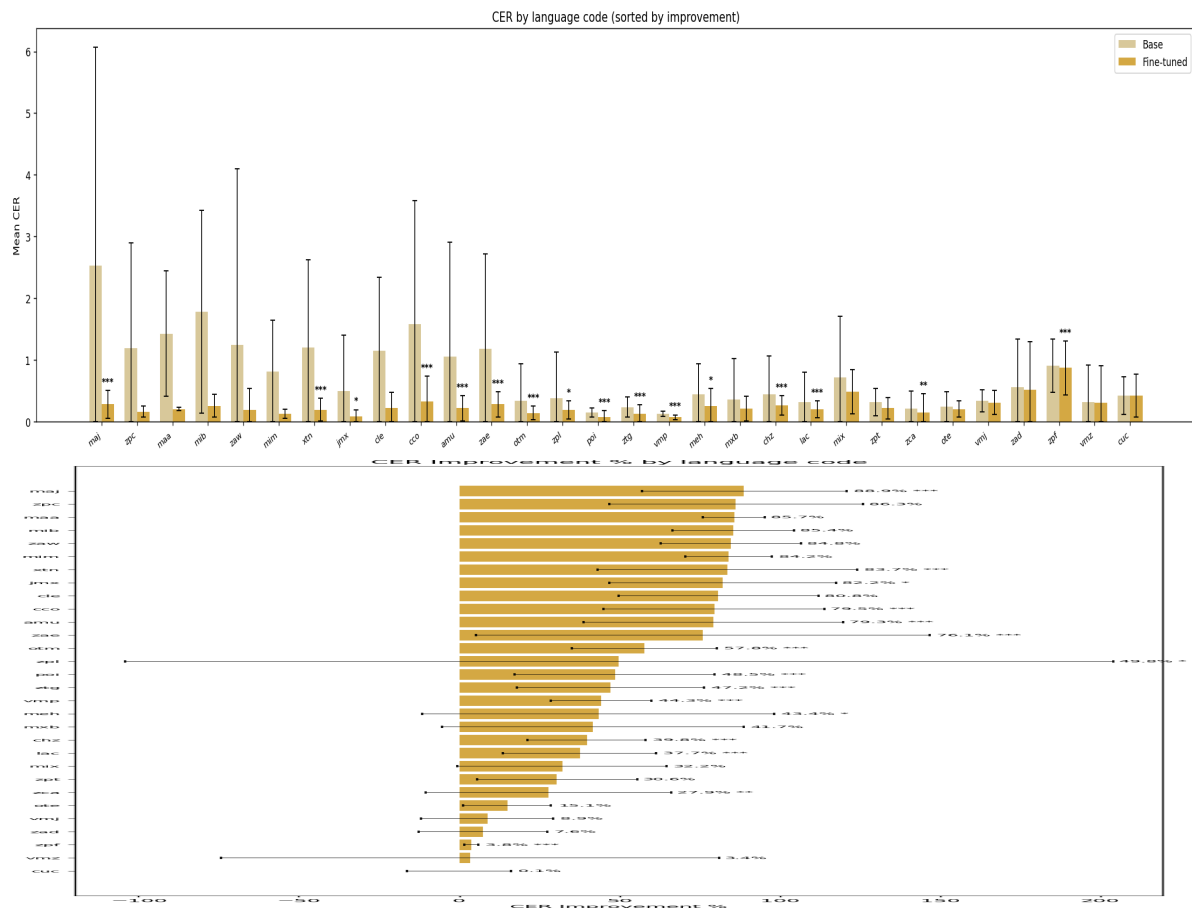
Score range	N pages	% of subset	Base CER	Fine-tuned CER	Reduction
[0.3, 0.4)	351	17.6%	0.793	0.241	−70%
[0.4, 0.5)	316	15.8%	0.833	0.235	−72%
[0.5, 0.6)	280	14.0%	1.010	0.245	−76%
[0.6, 0.7)	424	21.2%	0.542	0.198	−63%
[0.7, 0.8)	260	13.0%	0.433	0.190	−56%
[0.8, 0.9)	106	5.3%	0.786	0.252	−68%
[0.9, 1.0)	242	12.1%	1.065	0.292	−73%
Total	2,000	100%	0.773	0.232	−70%





4.5 Per-language results

The chart below shows CER by language code (sorted by improvement, descending). Languages with larger sample sizes (e.g., Amuzgo 328 pages, Chinanteco 217 pages) show the most reliable improvement estimates. Low-N languages (7 pages) often lack statistical significance despite large point estimates.



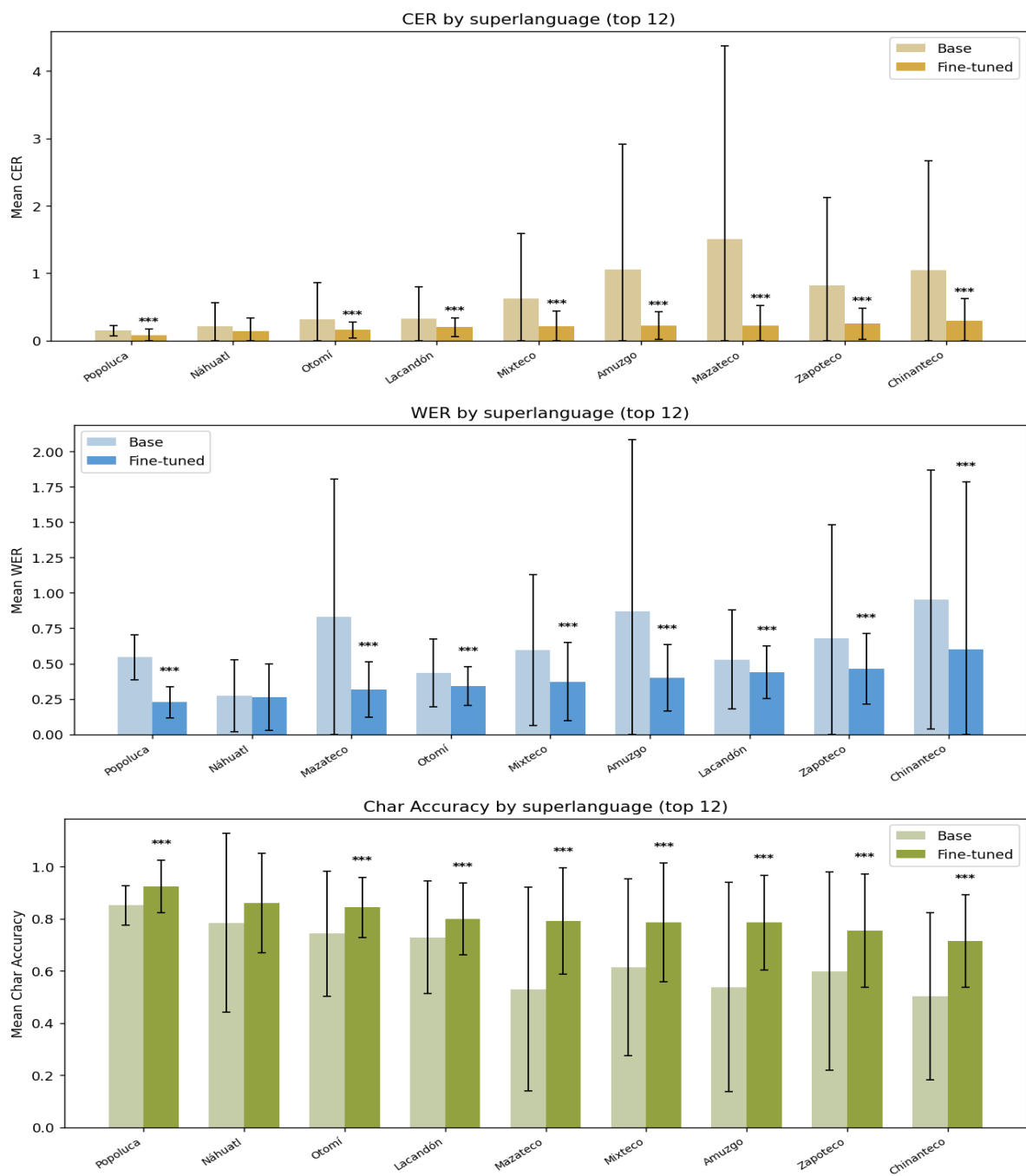
The following table lists the largest languages by page count (20 pages). Full results for all 30 languages are in the repository.

Cod e	Language	Pag es	Base CER	FT CER	Improv ement	Sig.
amu	Amuzgo	328	1.054	0.219	−79%	***
lac	Lacandón	431	0.322	0.200	−38%	**
cco	Chinanteco	217	1.584	0.325	−79%	***
chz	Chinanteco	187	0.439	0.264	−40%	***
zae	Zapoteco	165	1.179	0.281	−76%	***
otm	Otomí	146	0.340	0.143	−58%	***
zpl	Zapoteco	76	0.380	0.190	−50%	*
maj	Mazateco	64	2.534	0.282	−89%	***

mxh	Mixteco	50	0.362	0.211	-42%	**
ote	Otomí	42	0.242	0.205	-15%	ns
ztg	Zapoteco	42	0.237	0.125	-47%	**
vmp	Mazateco	33	0.131	0.073	-44%	***
xtn	Mixteco	32	1.201	0.196	-84%	*
jmx	Mixteco	22	0.498	0.088	-82%	***

4.6 Per-superlanguage results

Grouping languages by superlanguage shows clear performance variation across all three metrics — CER, WER, and character accuracy.

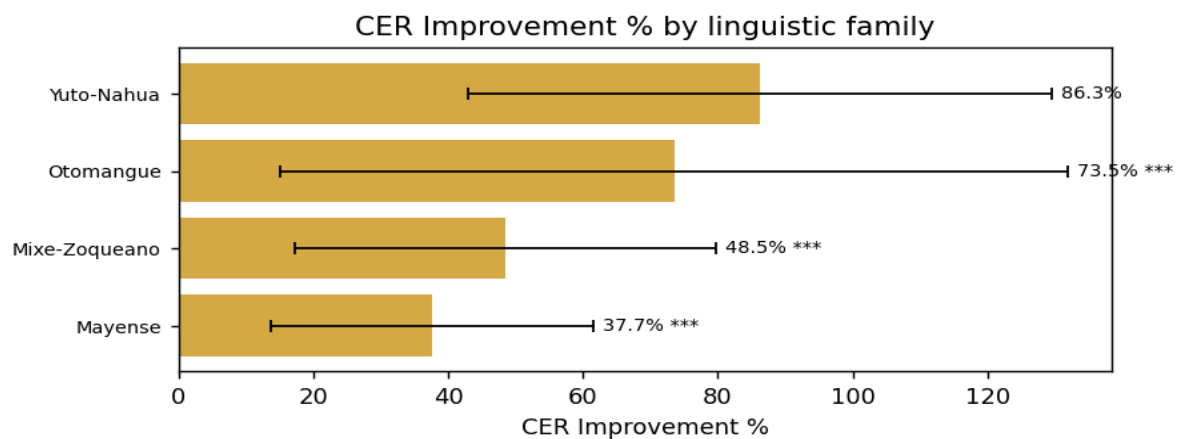
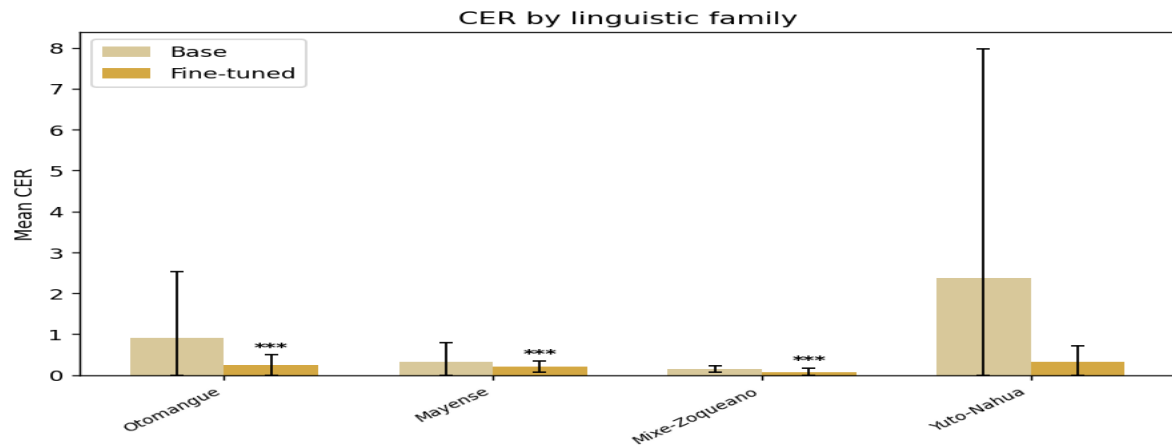


Superlanguage	Pages	Base CER	FT CER	Improvement	Sig.
Lacandón	431	0.322	0.200	-38%	**
Chinanteco	412	1.049	0.297	-72%	***
Amuzgo	328	1.054	0.219	-79%	***
Zapoteco	319	0.814	0.249	-69%	***

Otomí	188	0.318	0.157	−51%	***
Mixteco	146	0.620	0.214	−66%	***
Mazateco	117	1.505	0.224	−85%	***
Popoluca	16	0.149	0.077	−49%	***
Náhuatl	6	0.216	0.140	−35%	ns

4.7 Per-family results

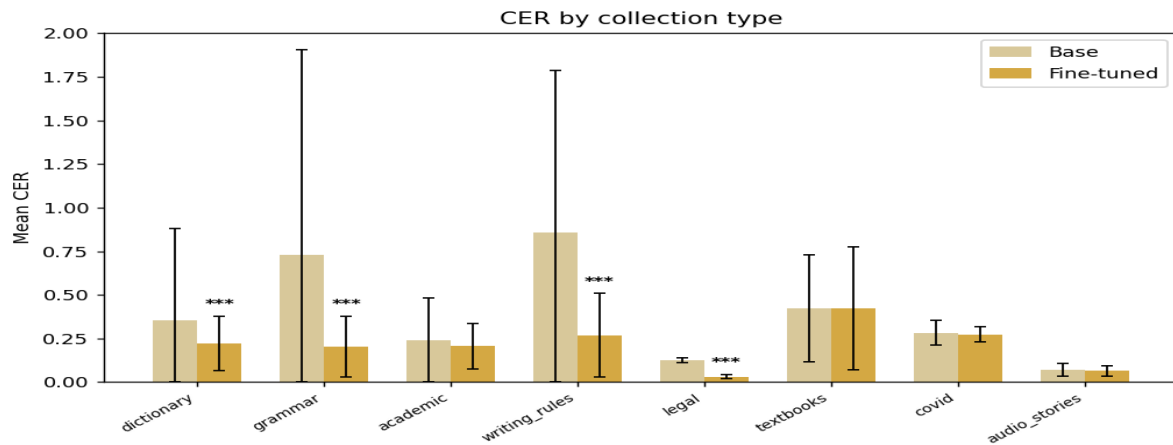
At the language family level, Otomangue (1,512 pages, 73.5% reduction) dominates the evaluation set as the largest family. Yuto-Nahua shows the most dramatic improvement (86.3%) but is limited by its small sample size (8 pages, ns).



Family	Pages	Base CER	FT CER	Improvement	Sig.
Otomangue	1512	0.902	0.239	-74%	***
Mayense	431	0.322	0.200	-38%	**
Mixe-Zoqueano	16	0.149	0.077	-49%	***
Yuto-Nahua	8	2.383	0.325	-86%	ns

4.8 Per-collection results

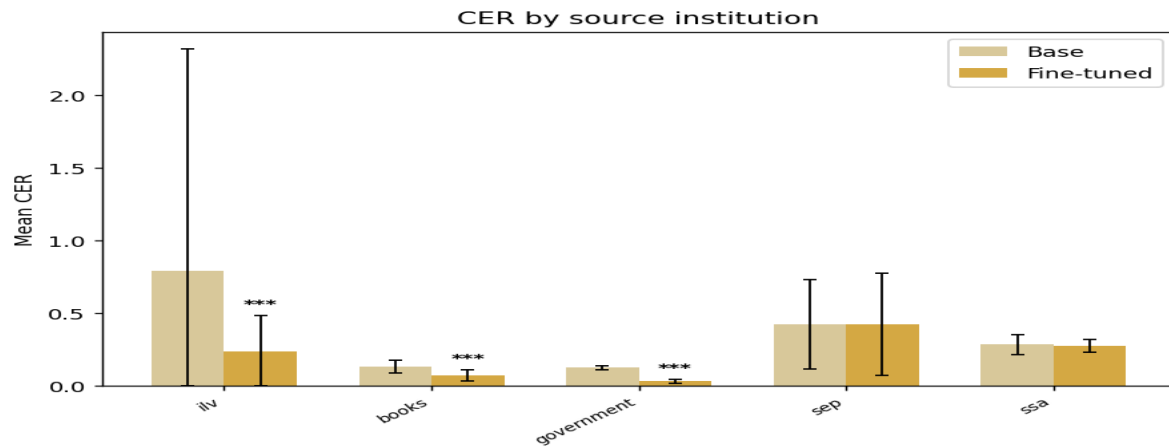
Dictionaries (643 pages) and grammars (632 pages) form the bulk of the evaluation set, both showing strong improvements (−37% and −72% respectively). Legal documents show the lowest absolute CER after fine-tuning (0.032) despite their small sample.



Collection	Pages	Base CER	FT CER	Improvement	Sig.
dictionary	643	0.357	0.223	−37%	***
grammar	632	0.730	0.203	−72%	***
academic	44	0.241	0.206	−15%	ns
writing_rules	35	0.856	0.269	−69%	***
legal	12	0.124	0.032	−75%	***
textbooks	4	0.424	0.423	−0%	ns
covid	3	0.283	0.274	−3%	ns
audio_stories	3	0.070	0.065	−7%	ns

4.9 Per-source results

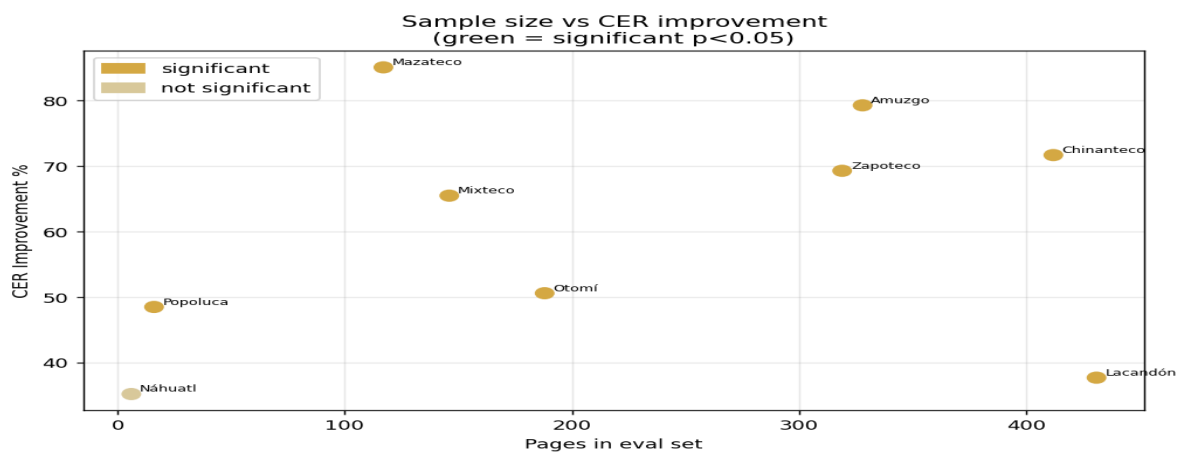
The vast majority of pages (1,937 of 2,000) come from the ILV (Instituto Lingüístico de Verano) source, reflecting the primary institutional source of indigenous language documentation in Mexico. Government and books sources show excellent post-fine-tuning CER.



Source	Pages	Base CER	FT CER	Improvement	Sig.
ilv	1937	0.793	0.236	-70%	***
books	33	0.131	0.073	-44%	***
government	12	0.124	0.032	-75%	***
sep	4	0.424	0.423	-0%	ns
ssa	3	0.283	0.274	-3%	ns

4.10 Key findings

- Catastrophic base model failure on low-score pages: The base model exceeds CER 1.0 in the [0.3–0.6) score ranges — meaning it inserts more spurious characters than the reference contains. This confirms the out-of-distribution failure hypothesis: indigenous-specific diacritics and tonal marks are absent from the base model's training distribution.
- Consistent improvement across all score ranges: The fine-tuned model brings every score range below CER 0.30, demonstrating that fine-tuning generalizes across the full spectrum of indigenous character density.
- Largest gains where they matter most: The most dramatic relative improvements occur in the [0.3–0.6) ranges (–70% to –76%), precisely where indigenous character density is highest and the base model fails most severely.
- Statistical robustness: All four metrics reach Wilcoxon $p < 0.0001$ on $n=2,000$ paired samples. The improvements are not attributable to random variation.
- WER vs CER: WER improvement (+38% relative) is more conservative than CER (+70%) due to the agglutinative morphology of many indigenous languages, where a single missed character can constitute a full word error. CER is the primary metric for this reason.
- Full results available: Per-item JSON results and per-language breakdowns are in the `evaluation/test_2000/` directory of the GitHub repository.



5. Open-Source Assets

All project assets are publicly available. Training data is not required to be open-sourced per the competition rules, but has been made public to maximize community value.

Asset	Location	Public?
Fine-tuned model	huggingface.co/tachiwin/Tachiwin-OCR-1.5	YES
Training dataset	huggingface.co/datasets/tachiwin/multilingual_ocr_llm_2	YES
Evaluation dataset	huggingface.co/datasets/tachiwin/ocr-test-challenging-3	YES
Demo — Document OCR	huggingface.co/spaces/tachiwin/document-ocr	YES
Demo — Multilingual OCR	huggingface.co/spaces/tachiwin/multilingual_ocr	YES
GitHub repository	github.com/tachiwin/tachiwin-ocr	YES
Evaluation script	Attached to submission email	YES

Both Hugging Face Spaces demos are live. Reviewers can test the model directly by uploading any document image at huggingface.co/spaces/tachiwin/document-ocr — no local setup required.